

Hyperkalemia — drugs (the saboteur workshop)

'THE DRUG SABOTEUR WORKSHOP'

Single underground industrial workshop, where medications that cause hyperkalemia are detailed as a distinct sabotage bays.

1 BAY #1: THE RAAS SABOTEURS

RAAS LANE



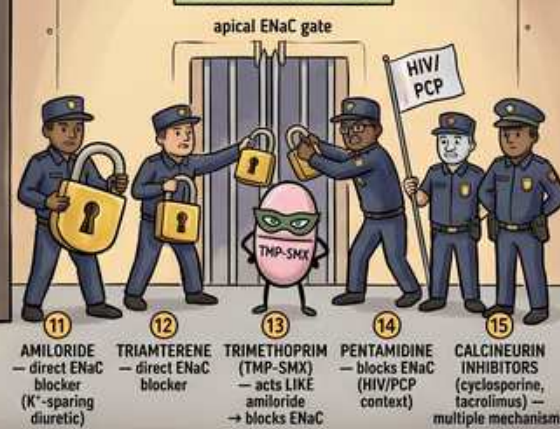
2 BAY #2: THE THRONE BLOCKERS

Aldo Receptor Blocker Lane



3 BAY #3: THE GATE LOCKERS

ENaC Blocker Lane



5 THE THREE-LANE PEARL SCROLL

① RAAS LANE

① RAAS LANE → ACEi, ARB, renin inhibitor, NSAID, heparin

② ALDO RECEPTOR

② **ALDO RECEPTOR** →
spironolactone, eplerenone,
drospirenone

③ ENaC GATE

③ ENaC GATE →
amiloride, triamterene,
TMP, pentamidine, CNI

**ALWAYS REVIEW
MEDS in any
new hyperK case**

REMEMBER: TMP ACTS LIKE AMILORIDE. NSAIDs ACT LIKE A RAAS BLOCKER. HEPARIN BLOCKS ALDO DIRECTLY.

1. 3 lanes of drugs

WHAT YOU SEE

The DRUG SABOTEUR WORKSHOP title over three sabotage bays (RAAS lane, Aldo-receptor lane, ENaC gate) — the three bays ARE the three mechanistic lanes that converge on impaired distal K⁺ secretion.

WHAT IT ENCODES

Drugs cause hyperkalemia along three mechanistic lanes that all converge on impaired distal K⁺ secretion: (1) RAAS BLOCKADE (less aldosterone is made), (2) ALDOSTERONE-RECEPTOR (mineralocorticoid receptor) blockade, and (3) ENaC blockade in the collecting duct. A few additional drugs cause a transcellular SHIFT (beta-blockers, digoxin).

BOARD TRAP

WRONG answer = assume every drug-induced hyperkalemia is from an ACEi/ARB. Map the drug to its lane (RAAS vs MR vs ENaC vs shift) — and remember risk multiplies when agents from different lanes are combined, especially in CKD or diabetes.

2. ACE inhibitors

WHAT YOU SEE

In Bay #1 (RAAS Saboteurs), the ACE-INHIBITOR saboteur cutting the angiotensin chain — severing the chain pictures less angiotensin II → less aldosterone → K⁺ retention.

WHAT IT ENCODES

ACE inhibitors (-pril) block angiotensin-converting enzyme → less angiotensin II → less aldosterone → reduced distal K⁺ secretion → hyperkalemia. Risk is highest in CKD, diabetes, volume depletion, or when combined with other RAAS/ENaC blockers. Monitor K⁺ and creatinine ~1–2 weeks after starting or uptitrating.

BOARD TRAP

WRONG answer = stop an ACEi permanently for a small, expected creatinine rise (<30%) without checking potassium. In proteinuric CKD the renoprotective benefit is worth tolerating mild Cr/K⁺ rises; severe or progressive hyperkalemia, however, does warrant dose reduction/discontinuation.

3. ARBs

WHAT YOU SEE

The ARB saboteur smashing the AT1 RECEPTOR in Bay #1 — blocking the receptor pictures the same drop in aldosterone, but downstream of the chain.

WHAT IT ENCODES

ARBs (-sartan) block the angiotensin II type-1 (AT1) receptor → less aldosterone → hyperkalemia. Same downstream effect as ACEi but without bradykinin accumulation (so no cough/angioedema risk).

BOARD TRAP

WRONG answer = switch an ACEi to an ARB to avoid hyperkalemia. The hyperkalemia risk is essentially the SAME because both reduce aldosterone; ARBs only fix the bradykinin-related cough/angioedema, not the potassium. Never combine an ACEi WITH an ARB (additive hyperkalemia/AKI, no benefit).

4. Renin inhibitor

WHAT YOU SEE

The aliskiren saboteur jamming RENIN at the very top of the RAAS pipe in Bay #1 — sabotaging the first step shuts the whole cascade.

WHAT IT ENCODES

Direct renin inhibitor (aliskiren) blocks renin at the top of the RAAS cascade → less angiotensin II and aldosterone → hyperkalemia.

BOARD TRAP

WRONG answer = add aliskiren to an ACEi or ARB in a diabetic/CKD patient. The ALTITUDE trial showed dual RAAS blockade with aliskiren caused excess hyperkalemia, hypotension, and renal events — it is contraindicated in diabetes with an ACEi/ARB.

5. NSAIDs

WHAT YOU SEE

The NSAID saboteur choking off PGE2 → renin in Bay #1 — cutting the prostaglandin signal pictures hyporeninemic hypoaldosteronism.

WHAT IT ENCODES

NSAIDs inhibit prostaglandin (PGE2) synthesis. PGE2 normally stimulates renin, so blocking it → low renin → low aldosterone (hyporeninemic hypoaldosteronism) → hyperkalemia. NSAIDs also reduce renal blood flow/GFR, compounding the effect.

BOARD TRAP

WRONG answer = overlook OTC NSAIDs as a hyperkalemia (and HTN/AKI) cause. NSAIDs act like a functional RAAS blocker; in a patient already on an ACEi + diuretic, adding an NSAID forms the dangerous 'triple whammy' (ACEi/ARB + diuretic + NSAID → AKI and hyperkalemia).

6. Heparin / LMWH

WHAT YOU SEE

The heparin saboteur attacking the adrenal aldosterone-synthesis machinery directly in Bay #1 — striking the gland itself, not a receptor.

WHAT IT ENCODES

Heparin (unfractionated and LMWH) directly inhibits aldosterone synthesis in the adrenal zona glomerulosa (reduces angiotensin-II receptor number and inhibits the final synthetic steps) → hyperkalemia, even at prophylactic doses and within days.

BOARD TRAP

WRONG answer = exclude heparin as a cause because 'it's just an anticoagulant' or because the dose is low. Heparin blocks aldosterone DIRECTLY at the adrenal (not at a receptor) and can cause hyperkalemia at any dose — a classic, easily missed cause in hospitalized patients.

7. Spironolactone

WHAT YOU SEE

In Bay #2 (Throne Blockers), the spironolactone wrestler body-blocking the mineralocorticoid (aldosterone) RECEPTOR throne — pinning the receptor blocks aldosterone's action.

WHAT IT ENCODES

Spironolactone is a non-selective mineralocorticoid (aldosterone) receptor antagonist in the collecting duct → blocks aldosterone action → K⁺-sparing → hyperkalemia. Used in HFrEF (RALES), resistant HTN (PATHWAY-2), primary aldosteronism, and cirrhotic ascites.

BOARD TRAP

WRONG answer = combine spironolactone with an ACEi/ARB in a patient with CKD or diabetes without close monitoring. This is a high-risk combination for severe hyperkalemia. Spironolactone's non-selective action also causes gynecomastia — switch to eplerenone if that occurs.

8. Eplerenone

WHAT YOU SEE

The eplerenone figure in Bay #2 selectively blocking the same MR throne, drawn cleaner (no antiandrogen baggage) — selectivity spares sex-hormone effects but keeps the hyperkalemia.

WHAT IT ENCODES

Eplerenone is a SELECTIVE mineralocorticoid receptor antagonist → blocks aldosterone with far fewer antiandrogen/progestin side effects (less gynecomastia) than spironolactone, but the same hyperkalemia risk.

BOARD TRAP

WRONG answer = assume eplerenone is 'safer' for potassium because it is selective. Selectivity only spares the sex-hormone side effects; the hyperkalemia risk is comparable to spironolactone and still requires K⁺ monitoring.

9. Drospirenone

WHAT YOU SEE

The drospirenone OCP pill in Bay #2 with spironolactone-like antimineralocorticoid action — a hidden hormonal-pill blocker of the same throne.

WHAT IT ENCODES

Drospirenone (the progestin in some combined oral contraceptives, e.g. Yaz/Yasmin) has antimineralocorticoid (spironolactone-like) activity → can raise serum K⁺.

BOARD TRAP

WRONG answer = forget the OCP when evaluating hyperkalemia in a young woman, especially if she is also on an ACEi/ARB, NSAID, or K-sparing diuretic. Drospirenone-containing pills carry a label caution about combining with potassium-raising drugs.

10. Amiloride/triamterene

WHAT YOU SEE

In Bay #3 (Gate Lockers), amiloride/triamterene padlocking the apical ENaC gate in the collecting duct — the locked Na⁺ channel removes the gradient that drives K⁺ secretion (works independent of aldosterone).

WHAT IT ENCODES

Amiloride and triamterene are K⁺-sparing diuretics that directly block the epithelial sodium channel (ENaC) in the collecting duct principal cells. Blocking Na⁺ reabsorption removes the favorable lumen-negative gradient that drives K⁺ secretion → hyperkalemia. They act independently of aldosterone (unlike spironolactone).

BOARD TRAP

WRONG answer = treat amiloride-induced hyperkalemia with fludrocortisone. ENaC blockade is downstream of aldosterone, so giving more mineralocorticoid won't help — stop the drug. Note: amiloride/triamterene block the channel directly (vs spironolactone, which blocks the receptor).

11. Trimethoprim

WHAT YOU SEE

The TMP figure slapping another padlock on the ENaC gate in Bay #3 — trimethoprim blocks ENaC exactly like amiloride.

WHAT IT ENCODES

Trimethoprim (in TMP-SMX/Bactrim) blocks ENaC exactly like amiloride → hyperkalemia, especially at high doses (PCP treatment), in CKD, or with other RAAS/K-sparing agents. Effect appears within days of starting.

BOARD TRAP

WRONG answer = miss TMP-SMX as the cause of new hyperkalemia in a patient treated for a UTI, cellulitis, or PCP. Trimethoprim is a classic, frequently tested ENaC blocker — think of it (and pentamidine) whenever an HIV/PCP patient becomes hyperkalemic.

12. Pentamidine

WHAT YOU SEE

The pentamidine figure locking the ENaC gate alongside TMP in Bay #3 — same ENaC block, the PCP-context culprit.

WHAT IT ENCODES

Pentamidine (used for PCP prophylaxis/treatment when TMP-SMX cannot be used) blocks ENaC like amiloride → hyperkalemia. Classic in the HIV/PCP context; also causes nephrotoxicity and hypoglycemia.

BOARD TRAP

WRONG answer = assume an HIV patient's hyperkalemia is only from their renal disease while overlooking pentamidine (or TMP-SMX). Both ENaC-blocking antimicrobials are common culprits in this exact population.

13. Calcineurin inhibitors

WHAT YOU SEE

The cyclosporine/tacrolimus saboteurs at the edge of Bay #3 hitting several distal targets at once — multi-mechanism transplant culprits causing a type-4-RTA-like state.

WHAT IT ENCODES

Calcineurin inhibitors (cyclosporine, tacrolimus) cause hyperkalemia by multiple distal mechanisms: they impair aldosterone responsiveness, downregulate ENaC and the Na-K-ATPase, and can produce a type-4-RTA-like state — a common problem in transplant recipients.

BOARD TRAP

WRONG answer = attribute a transplant patient's hyperkalemia solely to allograft dysfunction. CNIs directly impair distal K⁺ handling even with adequate graft function; review tacrolimus/cyclosporine levels and co-prescribed RAAS/ENaC blockers.

14. Deadly combos

WHAT YOU SEE

The DEADLY COMBINATIONS RACK at top right pairing bottles (ACEi/ARB + spironolactone, ACEi/ARB + NSAID, TMP-SMX + K-sparer) — stacking lanes multiplies the hyperkalemia risk.

WHAT IT ENCODES

High-risk drug combinations that multiply hyperkalemia risk: ACEi/ARB + spironolactone; ACEi/ARB + NSAID (+ diuretic = the 'triple whammy' for AKI); and TMP-SMX + a K-sparing diuretic or RAAS blocker. Risk compounds further in CKD, diabetes, and the elderly.

BOARD TRAP

WRONG answer = add a second potassium-raising drug from a different lane without rechecking potassium and renal function. Always reconcile the full medication list (including OTC NSAIDs and OCPs) — combinations from different lanes are additive and are the most common cause of severe drug-induced hyperkalemia.